A Derivation of the Black-Scholes Equation

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Math 4430X

March 26th 2027

1 Introduction

In this note, we derive the Black-Scholes equation using stochastic calculus. We assume that the underlying asset price follows a geometric Brownian motion, and the price of a derivative is given by a sufficiently smooth function of the asset price and time.

The key idea is to transform a stochastic system into a deterministic one. By applying Itô's formula to the derivative price and constructing a suitable hedging portfolio, we eliminate the randomness arising from Brownian motion. This leads to a locally risk-free portfolio, whose return must match the risk-free rate. From this condition, we obtain the Black-Scholes partial differential equation.

The derivation is organized into three steps: application of Itô's formula, construction of a self-financing portfolio, and elimination of the stochastic term.

2 Tools

Suppose S_t (stock price) follows the geometry Brownian Motion define by:

$$dS_t = \mu S_t dt + \sigma S_t dB_t$$

where σ = standard deviation, μ = drift coefficient, B_t = Standard Brownian motion, V(s,t) = Prices of financial derivatives
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Theorem 1 (One-dimensional Itô formula).
Suppose that $\phi = \phi(t,x) \in C^{1,2}([0,\infty) \times \mathbb{R}; \mathbb{R})$ and X_t is as in (12.4). Then

$$d\phi(t, X_t) = \phi_t(t, X_t)\, dt + \phi_x(t, X_t)\, dX_t + \frac{1}{2}\phi_{xx}(t, X_t)\, dX_t\, dX_t. \tag{1}$$

Theorem 2 (Itô product rule). Suppose that X_t and Y_t are Itô processes of the form

$$dX_t = a_t\, dt + b_t\, dB_t, \quad dY_t = c_t\, dt + d_t\, dB_t.$$

Then

$$d(X_t Y_t) = X_t\, dY_t + Y_t\, dX_t + dX_t\, dY_t. \tag{2}$$

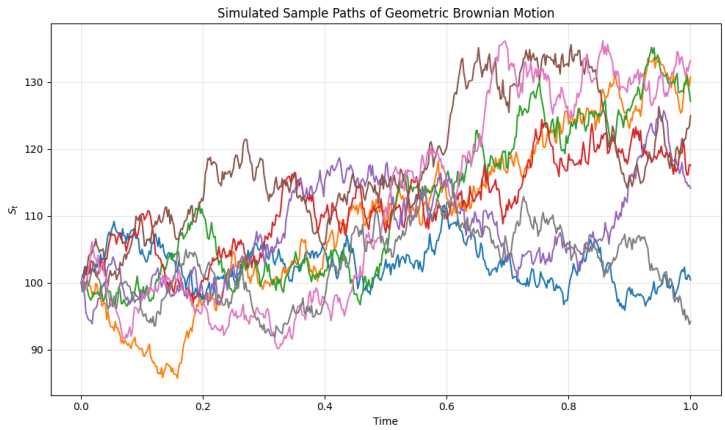

Figure 1: Simulated sample paths of geometric Brownian motion.

3 Derivation of the Black-Scholes Equation

Step 1 (Apply formula)

Apply Itô formula to $V(S_t, t)$, then we have:

$$dV = V_t dt + V_s dS_t + \frac{1}{2} V_{ss} (dS_t)^2 \tag{3}$$

Now plug the geometry Brownian motion into (3), Obtain :

$$dV = V_t dt + V_s(\mu S_t dt + \sigma S_t dB_t) + \frac{1}{2} V_{ss} (dS)^2 \tag{4}$$

Note that :

$$(dS)^2 = (\mu S_t dt + \sigma S_t dB_t)^2 = \mu^2 S_t^2 (dt)^2 + 2\mu_t^2 (dt dB_t) + \sigma^2 S_t^2 (dB_t)^2$$

By Ito multiplication table:

	dt	dB_t
dt	0	0
dB_t	0	dt

$$(dS)^2 = \mu^2 S_t^2 (0) + 2\mu_t^2 (0) + \sigma^2 S_t^2 (dt) = 0 + 0 + \sigma^2 S_t^2 dt = \sigma^2 S_t^2 dt$$

Now plug back to (4) obtain:

$$dV = V_t dt + V_s(\mu S_t dt + \sigma S_t dB_t) + \frac{1}{2} V_{ss} (\sigma^2 S_t^2 dt)$$

Rearrange the term for dt and dB_t we get:

$$dV = (V_t + V_s \mu S_t + \frac{1}{2}\sigma^2 S_t^2 V_{ss})dt + \sigma S_t V_s dB_t \tag{5}$$

where V is the value of the derivative and Δ represents the number of units of the underlying asset held in a short position.

Step 2 (Construct hedging portfoillo)

To proceed further, we aim to eliminate the randomness in the dynamics of the derivative price. The differential dV contains a stochastic term involving dB_t, which makes it uncertain. To overcome this, we construct a portfolio.

$$\Pi = V - \Delta S_t,$$

where V is the value of the derivative and Δ represents the number of units of the underlying asset held in a short position. By taking the differential of the portfolio,

$$d\Pi = dV - \Delta\, dS_t, \tag{6}$$

we can choose Δ appropriately so that the stochastic term involving dB_t cancels out. This allows us to obtain a locally risk-free portfolio whose evolution depends only on dt. This step is crucial, as it enables us to relate the portfolio's return to the risk-free rate and ultimately derive the Black-Scholes equation.

Proposition.

$$d\Pi = dV - \Delta\, dS_t$$

Proof: Note that

$$d\Pi = d(V - \Delta S_t),$$

By Theorem 2 we have,

$$d\Pi = dV - (\Delta S_t + S_t d\Delta + d\Delta dS_t),$$

$$d\Pi = dV - \Delta S_t - (S_t d\Delta + d\Delta dS_t),$$

Since Π is a self-financing portfolio, then

$$(S_t d\Delta + d\Delta dS_t) = 0$$

Hence,

$$d\Pi = dV - \Delta\, dS_t$$

□

Step 3 (Eliminate Brownian motion)

Plug (5) and geometry Brownian motion(dS_t) into (6), we have :

$$d\Pi = (V_t + V_s \mu S_t + \frac{1}{2}\sigma^2 S_t^2 V_{ss})dt + \sigma S_t V_s dB_t - \Delta(\mu S_t dt + \sigma S_t dB_t)$$

Rearrange it, we get:

$$d\Pi = (V_t + V_s \mu S_t + \frac{1}{2}\sigma^2 S_t^2 V_{ss} - \Delta \mu S_t)dt + (\sigma S_t V_s - \Delta S_t \sigma)B_t \tag{7}$$

Choice $\Delta = V_s$ to cencel dB_t

$$d\Pi = (V_t + V_s \mu S_t + \frac{1}{2}\sigma^2 S_t^2 V_{ss} - V_s \mu S_t)dt + (\sigma S_t V_s - \sigma S_t V_s)B_t$$

$$d\Pi = (V_t + \frac{1}{2}\sigma^2 S_t^2 V_{ss})dt \tag{8}$$

Note that

$$\Pi = V - \Delta S_t = V - V_s S$$

Since the portfolio Π has no stochastic term after hedging, it becomes locally risk-free. In a no-arbitrage market, any risk-free portfolio must earn the risk-free interest rate r. Therefore, its return satisfies

$$d\Pi = r\Pi\, dt.$$

Hence

$$d\Pi = r(V - V_s S)dt \tag{9}$$

Let (8)=(9)

$$r(V - V_s S)dt = (V_t + \frac{1}{2}\sigma^2 S_t^2 V_{ss})dt$$

Rearrange it, we obtain: **The Black-Scholes Equation**

$$V_t + \frac{1}{2}\sigma^2 S_t^2 V_{ss} - rV + rV_s S = 0$$

4 Additional Proof of Itô multiplication table

Proposition. $(dB_t)^2 = dt$.

Proof: Let B_t be a standard Brownian motion. For every fixed $t > 0$, it is sufficient to show that

$$\lim_{n\to\infty} \sum_{k=1}^{2^n} \left[B\left(\frac{k}{2^n}t\right) - B\left(\frac{k-1}{2^n}t\right) \right]^2 = t.$$

Let

$$\Delta_{nk} = B\left(\frac{k}{2^n}t\right) - B\left(\frac{k-1}{2^n}t\right), \quad k = 1, 2, \ldots, 2^n,$$

and define

$$X_{nk} = \Delta_{nk}^2 - \frac{t}{2^n}.$$

Therefore, to show

$$\sum_{k=1}^{2^n} \Delta_{nk}^2 \to t$$

is equivalent to showing

$$\sum_{k=1}^{2^n} X_{nk} \to 0.$$

For each n, the random variables $\{X_{nk} : k = 1, 2, \ldots, 2^n\}$ are independent and identically distributed. Moreover,

$$\mathbb{E}[X_{nk}] = \mathbb{E}[\Delta_{nk}^2] - \frac{t}{2^n} = 0.$$

Next, compute the second moment:

$$\mathbb{E}[X_{nk}^2] = \mathbb{E}\left[\left(\Delta_{nk}^2 - \frac{t}{2^n}\right)^2\right] = \mathbb{E}\left[\Delta_{nk}^4 + \frac{t^2}{4^n} - 2\frac{t}{2^n}\Delta_{nk}^2\right].$$

Since $\Delta_{nk} \sim \mathcal{N}(0, t/2^n)$, we have

$$\mathbb{E}[\Delta_{nk}^2] = \frac{t}{2^n}, \quad \mathbb{E}[\Delta_{nk}^4] = 3\left(\frac{t}{2^n}\right)^2.$$

Thus,

$$\mathbb{E}[X_{nk}^2] = 3\frac{t^2}{4^n} + \frac{t^2}{4^n} - 2\frac{t}{2^n} \cdot \frac{t}{2^n} = \frac{2t^2}{4^n}.$$

Since $\mathbb{E}[X_{nk}X_{nj}] = 0$ for $j \neq k$, we obtain

$$\mathbb{E}\left[\left(\sum_{k=1}^{2^n} X_{nk}\right)^2\right] = \sum_{k=1}^{2^n} \mathbb{E}[X_{nk}^2] = 2^n \cdot \frac{2t^2}{4^n} = \frac{2t^2}{2^n} \to 0.$$

as $n \to \infty$. This shows that

$$\sum_{k=1}^{2^n} X_{nk} \to 0$$

in mean square sense. □

Proposition. $(dB_t)(dt) = 0$.

Proof: Since $(dB_t)^2 = dt$ has already been established, we formally write

$$(dB_t)(dt) = (dB_t)^3.$$

Let B_t be a standard Brownian motion. For every fixed $t > 0$, it is sufficient to show that

$$\lim_{n\to\infty} \sum_{k=1}^{2^n} \left[B\left(\frac{k}{2^n}t\right) - B\left(\frac{k-1}{2^n}t\right)\right]^3 = 0.$$

Let

$$\Delta_{nk} = B\left(\frac{k}{2^n}t\right) - B\left(\frac{k-1}{2^n}t\right), \quad k = 1, 2, \ldots, 2^n,$$

and define

$$X_{nk} = \Delta_{nk}^3.$$

Then $\{\Delta_{nk}\}$ are independent with

$$\Delta_{nk} \sim \mathcal{N}\left(0, \frac{t}{2^n}\right).$$

Hence,

$$\mathbb{E}[X_{nk}] = \mathbb{E}[\Delta_{nk}^3] = 0.$$

Next, compute the second moment:

$$\mathbb{E}[X_{nk}^2] = \mathbb{E}[\Delta_{nk}^6].$$

For a normal random variable $Z \sim \mathcal{N}(0, \sigma^2)$, we have

$$\mathbb{E}[Z^6] = 15\sigma^6.$$

Thus,

$$\mathbb{E}[X_{nk}^2] = 15\left(\frac{t}{2^n}\right)^3 = \frac{15t^3}{2^{3n}}.$$

Since $\mathbb{E}[X_{nk}X_{nj}] = 0$ for $j \neq k$, we obtain

$$\mathbb{E}\left[\left(\sum_{k=1}^{2^n} X_{nk}\right)^2\right] = \sum_{k=1}^{2^n} \mathbb{E}[X_{nk}^2] = 2^n \cdot \frac{15t^3}{2^{3n}} = \frac{15t^3}{2^{2n}} = \frac{15t^3}{4^n} \to 0.$$

as $n \to \infty$. Therefore,

$$\sum_{k=1}^{2^n} X_{nk} \to 0$$

in mean square sense. □

Some clues of the proof can be found in Chapter 7 of "A First Course in Stochastic Processes" written by Samuel Karlin and Howard M.Taylor [1] [2] and more details in [3].

5 Conclusion

In the derivation of the Black-Scholes equation, three key ideas appear.

First, we use the Itô multiplication table and the chain rule to simplify complicated expressions. Second, we construct a self-financing portfolio to eliminate the Brownian motion term. Third, we use the fact that in a no-arbitrage market, any risk-free portfolio must earn the risk-free interest rate r, which allows us to set (8) equal to (9) and obtain the final equation. This shows that the final result is not accidental, but comes from a chain of mathematical and financial arguments working together.